NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 3 CHRONOLOGICAL ENQUIRY SEQUENCE • C1700–C1900

Medicine in 18th & 19th Century Britain

c1700–c1900: Ideas on Causes, Prevention, Hospital Care, Surgery & Public Health



HISTORICAL PRIMARY RECORD • 1802 — HISTORICAL SATIRE ARCHIVE

James Gillray, The Cow-Pock—or—the Wonderful Effects of the New Inoculation! (1802)

Hand-coloured etching lampooning early Victorian fears that Edward Jenner's cowpox vaccine would cause patients to sprout bovine horns and calves.

PRIMARY VISUAL EVIDENCE — THE BRITISH MUSEUM DEPARTMENT OF PRINTS & DRAWINGS

PEARSON EDEXCEL OPTION 11 • KEY TOPIC 3 CONTENT — OFFICIAL SPECIFICATION DIRECTORY

1 Ideas About the Cause of Disease and Illness

- Continuity and change in explanations of the cause of disease and illness. The influence in Britain of **Pasteur's Germ Theory** and **Koch's work on microbes**.

🎯 **CAUSATION TURNING POINT**
From Spontaneous Generation to Germs: How Pasteur's 1861 swan-neck flask experiments and Koch's chemical dye staining isolated specific pathogens, decisively destroying miasma.

2 Approaches to Prevention and Treatment

- The extent of change in care and treatment: improvements in hospital care and the **influence of Nightingale** on nursing and hospitals in Britain. The impact of **anaesthetics and antiseptics** on surgery.
- New approaches to prevention: the development and use of vaccinations and the **Public Health Act (1875)**.

3 Case Studies

- Key individual: **Jenner and the development of vaccination**.
- Fighting **Cholera in London (1854)**; attempts to prevent its spread; the significance of **Snow and the Broad Street pump**.

EXAM MATRIX EDEXCEL PAPER 1 SECTION B EXAMINATION TARGETING FRAMEWORK • THEMATIC STUDY [36M + 4M SPAG • 55 MINS] — SPECIFICATION 1HI0/11

Q3 • 4 MARKS — AO1/AO2 CONCEPT DRILL

Similarity or Difference Across Eras

Target: Explain 1 way [ideas / treatments / care] in this era were similar or different to another period.
Rule: Provide specific factual evidence from *both* named periods to secure 4/4 marks (approx. 5 mins).

Practice: In-Workbook 4-Mark Comparative Tasks

Q4 • 12 MARKS — AO1/AO2 CAUSATION

Explain Why [Causal Change] Occurred

Target: Multi-causal analytical essay explaining why medical change or continuity took place.
Rule: Write 3 developed PEEL paragraphs. Must address both stimulus points + 1 distinct point of own knowledge (18 mins).

Practice: Chronological Inquiry Spine & Causation Drills

Q5/6 • 16+4 SPaG — AO1/AO2/AO4 SYNOPTIC

'How Far Do You Agree?' Evaluative Essay

Target: Choose Q5 OR Q6. Evaluative thematic essay assessing factor significance or change over time.
Rule: Balanced argument (for/against) with sustained criteria and a decisive historical verdict (30 mins).

Practice: Full 16-Mark Extended Writing Frames

Vaccination, Anaesthetics & Early Public Health Battles

REVISION MIND-MAP CANVAS: At the end of each lesson, use the app's *Consolidate Station* to retrieve your keywords and branch out your mind-map across the open page (Problems & Opposition to the left • Breakthroughs & Impact to the right).

◂ VARIOLATION RISK & OPPOSITION

1796 Edward Jenner & Smallpox Vaccine (KT 3.2)

COWPOX TRIAL & 1852 LAW ▸

◂ DOSAGE FEAR & BLACK PERIOD

1847 James Simpson & Chloroform (KT 3.4)

PAIN-FREE INVASIVE SURGERY ▸

◂ LAISSEZ-FAIRE & RATEPAYERS

1848 First Public Health Act (Permissive) (KT 3.5)

CHADWICK REPORT & VOLUNTARY ACT ▸

◂ SOHO CHOLERA & MIASMA DOUBT

1854 John Snow & Broad Street Pump (KT 3.5)

SPOT MAP & PUMP HANDLE PROOF ▸

SPECIFICATION SPINE CONTINUES TO GERM THEORY, LISTER & 1875 ACT (PAGE 3) ➔

The 18th century relied on empirical observation; the 19th century unlocked the laboratory.

Hospital Reform, Bacteriology, Antisepsis & Compulsory Public Health

REVISION MIND-MAP CANVAS: At the end of each lesson, use the app's *Consolidate Station* to retrieve your keywords and branch out your mind-map across the open page (Scientific Discovery to the left • Clinical & State Reform to the right).

◂ SCUTARI SQUALOR & MORTALITY

1854–59 Florence Nightingale & Nursing (KT 3.3)

PAVILION WARDS & NURSE SCHOOL ▸

◂ SPONTANEOUS GENERATION MYTH

1861 Louis Pasteur's Germ Theory (KT 3.1)

SWAN-NECK FLASKS & MICROBES ▸

◂ SEPSIS & SURGEON RESISTANCE

1865 Joseph Lister & Carbolic Spray (KT 3.4)

ANTISEPTICS TO ASEPTIC THEATRE ▸

◂ KOCH STAINS SPECIFIC BACTERIA

1875–82 1875 Health Act & Robert Koch (KT 3.5 • 3.1)

COMPULSORY SEWERS & CLEAN WATER ▸

KEY TOPIC 3 SPECIFICATION SPINE COMPLETE ➔

Why did it take 30 years for Germ Theory to transform medical understanding in Britain?

AUDIO • QUIZ

1 c1700–1850s • Why did doctors believe disease created germs, rather than vice versa?

- Spontaneous generation theory
- Miasma stench blamed
- Pre-bacteriology limits

2 1861 • How did the swan-neck flask disprove spontaneous generation?

- Louis Pasteur (1861)
- Swan-neck flask test
- Airborne microbes proved

3 1867–1876 • Why did British doctors refuse to believe germs caused human illness?

- British medical scepticism
- John Tyndall experiments
- Charlton Bastian opposition

4 1876–1882 • How did Koch's staining and culture methods prove specificity?

- Robert Koch bacteriology
- Agar jelly plates
- Methyl violet stains

5 1883–1890 • Why was photomicrography crucial in winning British doctors over?

- Tuberculosis bacteria (1882)
- Cholera microbe (1883)
- High-resolution microphotography

Pasteur proved that rotting soup doesn't spontaneously create germs; Koch proved which specific germs caused tuberculosis.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1860–c1890. [12 marks]

Stimulus: **Louis Pasteur's Germ Theory (1861)** **Robert Koch's work on bacteria** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT3.1 on Pages 2–3: map spontaneous generation and British scepticism (left) against Pasteur's Germ Theory and Koch's staining methods (right). **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which ideas about the cause of disease in the late nineteenth century (c1860–c1890) were different from ideas in the Renaissance (c1500–c1700). [4 marks]

Contrast laboratory discoveries of specific microbes with earlier beliefs in miasma and spontaneous generation.

Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?

AUDIO • QUIZ

1 1721–1795 • Why was inoculation dangerous for poor urban communities?

- Smallpox variolation inoculation
- Suttonian method fees
- Infection risk remains

2 1796–1798 • How did Jenner use scientific experimentation to test folklore?

- Edward Jenner (1796)
- James Phipps vaccination
- Cowpox blister pus

3 1798–1810s • Why did London doctors and church leaders resist vaccination?

- Royal Society scepticism
- Anti-vaccine cartoon satire
- Cowpox contamination fears

4 1802–1840 • Why did the British government abandon laissez-faire for vaccination?

- Parliamentary grants (£30,000)
- 1840 Vaccination Act
- Inoculation banned legally

5 1852–1871 • Why did making vaccination compulsory spark fierce public protests?

- Compulsory Vaccination (1853)
- 1871 Enforcement Act
- Dramatic smallpox eradication

Cartoons showed vaccinated people growing cow heads and horns: proof that anti-vax memes existed long before social media.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Edward Jenner's development of the smallpox vaccine was the most significant breakthrough in the prevention of disease in the period c1700–c1900.' How far do you agree? [16+4 marks]

Stimulus: **Inoculation** **The Public Health Act (1875)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT3.2 on Pages 2–3: map variolation risks and anti-vaccine opposition (left) against Jenner's 1796 cowpox trial and 1852 compulsory vaccination (right). **[ ] Done**

Jenner saved millions with cowpox, yet died before anyone could explain what a virus actually was.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of preventing smallpox in the late eighteenth century were different from methods used in the seventeenth century. [4 marks]

Contrast safe cowpox vaccination with earlier reliance on prayer, quarantine, and risky variolation.

How did Florence Nightingale transform hospitals from death traps into places of cure?

AUDIO • QUIZ

1 c1700–1850 • Why were early Victorian hospital wards notorious death traps?

- Pauper hospital squalor
- Untrained nurse reputation
- Cross-infection septic wards

2 1854–1856 • How did Nightingale's sanitary reforms drastically cut the death rate?

- Crimean War (1854)
- Scutari hospital sanitation
- Death rate plummeted (42% to 2%)

3 1859 • Why did Nightingale's belief in miasma still produce effective hygiene?

- Notes on Nursing (1859)
- Sanitary statistical charts
- Pavilion hospital architecture

4 1860s • How did the pavilion plan prevent cross-infection between wards?

- Separate pavilion wards
- Cross-ventilation high windows
- Tiled washable surfaces

5 1860–1890s • How did nurse training permanently raise the standard of hospital care?

- St Thomas' Training School
- Nightingale Fund (£44,000)
- Professional disciplined matrons

Nightingale believed miasma caused disease, yet her obsession with fresh air, scrubbing, and clean drains saved thousands of lives.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why the standard of hospital care and nursing improved so significantly in the second half of the nineteenth century. [12 marks]

Stimulus: **Florence Nightingale's work in the Crimea (1854–56)** | **The pavilion hospital plan** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT3.3 on Pages 2–3: map Scutari squalor and septic wards (left) against Nightingale's pavilion wards and St Thomas' training school (right). **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which hospital care in the late nineteenth century was different from hospital care in the Medieval period. [4 marks]

Contrast modern sanitary pavilion wards and trained medical care with medieval monastic care.

Why did the conquest of pain in surgery temporarily increase the patient death rate?

AUDIO • QUIZ

1 c1700–1846 • Why was surgical speed essential before the discovery of anaesthetics?

- Speed-focused surgery pain
- Surgical shock mortality
- Ether & nitrous oxide

2 1847–1853 • Why did the death of Hannah Greener cause panic among surgeons?

- James Simpson (1847)
- Chloroform anaesthetic vapour
- Queen Victoria childbirth (1853)

3 1846–1865 • Why did conquering surgical pain lead to higher patient death rates?

- 'Black Period' surgery
- Deeper invasive operations
- Hospital gangrene surge

4 1865–1870s • How did Lister apply Germ Theory to solve hospital gangrene?

- Joseph Lister (1865)
- Carbolic acid spray
- Compound fracture antisepsis

5 1880s–1890s • What is the crucial difference between antiseptic and aseptic surgery?

- Aseptic surgical theatre
- Autoclave steam sterilisation
- Sterile gloves & masks

Chloroform conquered pain, but Lister conquered the invisible enemy: the rotting microbes thriving on surgeons' unwashed hands.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why surgery became significantly safer in the period c1840–c1900. [12 marks]

Stimulus: **James Simpson's discovery of chloroform (1847)** | **Joseph Lister's use of carbolic acid (1865)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT3.4 on Pages 2–3: map the surgical 'Black Period' and sepsis (left) against Simpson's chloroform, Lister's carbolic spray, and aseptic theatres (right). **[ ] Done**

Surgeons used to take pride in blood-caked frock coats; Lister forced them to wash their hands and operate in carbolic acid mist.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which surgical operations in the early nineteenth century (before 1846) were similar to surgery in the Medieval period. [4 marks]

Focus on raw surgical speed and physical restraint before effective chemical anaesthetics were discovered.

Why did it take 20 years for John Snow's cholera discovery to change government policy?

AUDIO • QUIZ

1 1831–1848 • Why did towns refuse to set up health boards under the 1848 Act?

- Cholera epidemics (1831–1854)
- Laissez-faire government policy
- Permissive 1848 Health Act

2 1854 • How did Snow use epidemiological mapping to prove water transmission?

- John Snow Soho (1854)
- Broad Street spot map
- Pump handle removed

3 1854–1858 • Why did the medical establishment refuse to accept Snow's water theory?

- Water company comparison
- Southwark vs Lambeth
- Waterborne transmission proved

4 1858–1865 • Why did fear of miasma paradoxically save thousands from cholera?

- Great Stink (1858)
- Joseph Bazalgette sewers
- Underground London interceptors

5 1875 • Why was the 1875 Act the definitive death blow to laissez-faire?

- 1875 Public Health Act
- Compulsory town sewers
- End of laissez-faire

Snow removed the Broad Street pump handle in 1854, proving cholera was water-borne seven years before Germ Theory.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'The work of John Snow was the main reason for improvements in public health in the nineteenth century.' How far do you agree? [16+4 marks]

Stimulus: Edwin Chadwick's 1842 Report | The Public Health Act (1875) *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, a major breakthrough was... • For example, ... • This was significant because... • On the other hand, an alternative factor was... • In particular, ... • Overall, the most decisive factor was... because...*

Write your response in the space provided below: *(Response continues on facing page)*

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT3.5 on Pages 2–3: map laissez-faire and the 1848 Act's limits (left) against Snow's 1854 Broad Street proof, Bazalgette's sewers, and the 1875 Act (right). [ ] Done

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which government action on public health in the late nineteenth century was different from government action in the Renaissance. [4 marks]

Contrast compulsory public health legislation with earlier temporary emergency measures.

KEY TOPIC 3 NOTES & EXAMINATION STRATEGY

Notes: Record key causal links, essay plans, and teacher feedback for this unit.

PUPIL NAME: _____ **TARGET GRADE:** ☐

HW DATES & QR	ENQUIRY FOCUS & PARENTAL HOMEWORK VERIFICATION 1. Scan QR to read aloud / track audio narration • 2. Complete 10-Q online quiz • 3. Parent verifies & signs	DO NOW (10M) Topic Recall	Q3 SIM/DIFF	Q4/5 EXTENDED
SET: _____  DUE: _____	**KT3.1: Ideas on Causes: Pasteur's Germ Theory & Koch's Bacteriology** *"Why did it take 30 years for Germ Theory to transform medical understanding in Britain?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 3.1 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT3.2: Approaches to Prevention: Edward Jenner & The Smallpox Vaccine** *"Why did Edward Jenner face fierce resistance despite discovering the smallpox vaccine?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 3.2 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT3.3: Improvements in Hospital Care: Florence Nightingale & Professional Nursing** *"How did Florence Nightingale transform hospitals from death traps into places of cure?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 3.3 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT3.4: The Surgical Revolution: Simpson's Chloroform & Lister's Antiseptics** *"Why did the conquest of pain in surgery temporarily increase the patient death rate?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 3.4 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT3.5: Public Health & Cholera: John Snow & The 1875 Public Health Act** *"Why did it take 20 years for John Snow's cholera discovery to change government policy?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 3.5 RECALL** _____ **/ 10**	_____ **/ 4**	**Q5/6 Essay** _____ **/ 20**

Retain this student assessment record for parent consultations and GCSE revision monitoring.